



University of Pretoria

Semester Test 1
Semestertoets 1

FSK176

19 August/us 2009

Surname and Initials: _____ Student Number: _____

Van en voorletters:

Studentennummer:

The undersigned hereby confirms that he/she is familiar with and accepts the examination regulations.

Ondergetekende bevestig hiermee dat hy/sy op hoogte is met die eksamenregulasies en dat hy/sy hom/haar daaraan onderwerp.

Signature: _____

Handtekening:

Time : 2 hours

Tyd:

Total : 85 max

Totaal:

Answer all questions. Programmable calculators may NOT be used. No textbooks allowed. A formula sheet is attached at the end of this paper	<i>Beantwoord alle vrae.</i> <i>Programeerbare sakrekenaars MAG NIE gebruik word nie.</i> <i>Geen handboeke word toegelaat.</i> <i>'n Formuleblad is agter aan die vraestel aangeheg.</i>
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SEMESTER TEST 1	FSK 176	SEMESTERTOETS 1
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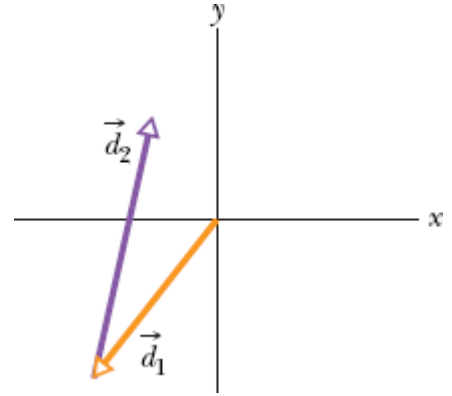
Venue:_____

Lokaal:

Question 1**[3]**

The two vectors shown in the Figure lie in an xy plane. What are the signs of the x and y components, respectively, of

- (a) $\vec{d}_1 + \vec{d}_2$,
 (b) $\vec{d}_1 - \vec{d}_2$ and
 (c) $\vec{d}_2 - \vec{d}_1$?



Twee vektore word aangedui in die Figuur. Hulle lê in die xy vlak. Wat is die tekens van die x en y komponente vir die volgende:

- (a) $\vec{d}_1 + \vec{d}_2$,
 (b) $\vec{d}_1 - \vec{d}_2$ and
 (c) $\vec{d}_2 - \vec{d}_1$?

(a) x component is negative and y component is positive ✓
(b) x component is negative and y component is negative ✓
(c) x component is positive and y component is positive ✓

Question 2**[3+3+1 = 7]**

Here are two vectors: $\vec{a} = (4,0 \text{ m})\hat{i} - (3,0 \text{ m})\hat{j}$ and $\vec{b} = (6,0 \text{ m})\hat{i} + (8,0 \text{ m})\hat{j}$.

- (a) What are the magnitude and the angle (relative to $\hat{i}$) of $\vec{b} - \vec{a}$?
 (b) What are the magnitude and the angle (relative to $\hat{i}$) of $\vec{a} - \vec{b}$?
 (c) What is the angle between the directions of $\vec{b} - \vec{a}$ and $\vec{a} - \vec{b}$?

Hier is twee vektore: $\vec{a} = (4,0 \text{ m})\hat{i} - (3,0 \text{ m})\hat{j}$ and $\vec{b} = (6,0 \text{ m})\hat{i} + (8,0 \text{ m})\hat{j}$.

- (a) Wat is die grootte en die hoek (relatief tot $\hat{i}$) van $\vec{b} - \vec{a}$?
 (b) Wat is die grootte en die hoek (relatief tot $\hat{i}$) van $\vec{a} - \vec{b}$?
 (c) Wat is die hoek tussen $\vec{b} - \vec{a}$ and $\vec{a} - \vec{b}$?

(a) $\vec{b} - \vec{a} = (6-4)\hat{i} + (8-(-3))\hat{j} = (2)\hat{i} + (11)\hat{j}$ ✓
$ \vec{b} - \vec{a} = \sqrt{(2)^2 + (11)^2} = 11.18 \text{ m}$ ✓
$\tan \theta = \frac{y}{x} = \frac{11}{2}$ ✓ $\theta = 79.70^\circ$
(b) $\vec{a} - \vec{b} = (4-6)\hat{i} + (-3-8)\hat{j} = (-2)\hat{i} + (-11)\hat{j}$ ✓
$ \vec{a} - \vec{b} = \sqrt{(-2)^2 + (-11)^2} = 11.18 \text{ m}$ ✓
$\tan \theta = \frac{y}{x} = \frac{-11}{-2} = 79.70^\circ$ $\theta = 79.70^\circ$ but both components are negative, therefore from the +ve x -axis the resultant is 259.70° (or -100.30°) ✓
(c) Angle between two resultant vectors is $259.7 - 79.7 = 180^\circ$ ✓

Question 3**[2+4+1 = 7]**

Vectors $\vec{A}$ and $\vec{B}$ lie in an xy plane. $\vec{A}$ has magnitude 8,00 and angle 130° ; $\vec{B}$ has components $B_x = -7,72$ and $B_y = -9,20$.

- (a) Determine $5\vec{A} \cdot \vec{B}$?
 (b) Determine $4\vec{A} \times 3\vec{B}$ in unit-vector notation.
 (c) What is the angle between the directions of $\vec{A}$ and $4\vec{A} \times 3\vec{B}$?

Vektore $\vec{A}$ en $\vec{B}$ lê in die xy vlak. $\vec{A}$ het 'n grootte van 8,00 en 'n hoek van 130° ; $\vec{B}$ het komponente $B_x = -7,72$ en $B_y = -9,20$.

- (a) Bereken $5\vec{A} \cdot \vec{B}$?
 (b) Bereken $4\vec{A} \times 3\vec{B}$ in eenheidsvektor notasie.
 (c) Wat is die hoek tussen die rigtings van $\vec{A}$ and $4\vec{A} \times 3\vec{B}$?

$$\vec{A} = 8.00(\cos 130^\circ \hat{i} + \sin 130^\circ \hat{j}) = -5.14 \hat{i} + 6.13 \hat{j} \quad \checkmark$$

$$\vec{B} = B_x \hat{i} + B_y \hat{j} = -7.72 \hat{i} - 9.20 \hat{j}. \quad \checkmark$$

$$(a) \quad 5\vec{A} \cdot \vec{B} = 5(-5.14 \hat{i} + 6.13 \hat{j}) \cdot (-7.72 \hat{i} - 9.20 \hat{j}) = 5[(-5.14)(-7.72) + (6.13)(-9.20)] \quad \checkmark \checkmark$$

$$= -83.4.$$

$$(b) \quad 4\vec{A} \times 3\vec{B} = 12\vec{A} \times \vec{B} = 12(-5.14 \hat{i} + 6.13 \hat{j}) \times (-7.72 \hat{i} - 9.20 \hat{j}) = 12(94.6 \hat{k}) = 1.14 \times 10^3 \hat{k} \quad \checkmark \checkmark$$

$$(c) \quad \text{Since } \vec{A} \text{ is in the } xy \text{ plane, and } \vec{A} \times \vec{B} \text{ is perpendicular to that plane, then the answer is } 90^\circ. \quad \checkmark$$

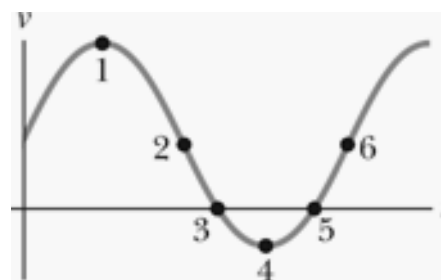
Question 4**[2+1+2+2 = 7]**

The Figure gives the velocity of a particle moving along an axis. Point 1 is at the highest point on the curve; point 4 is at the lowest point; and points 2 and 6 are at the same height.

- (a) Describe the motion of the particle at time $t = 0$?
 (b) What is the direction of travel at point 4?
 (c) At which of the six numbered points does the particle reverse its direction of travel? Give reason(s) for your answer.
 (d) At which of the six numbered points does the particle have the greatest magnitude of the acceleration? Give reason(s) for your answer.

Die Figuur dui die snelheid van 'n deeltjie aan wat langs 'n as beweeg. Punt 1 is die hoogste punt op die kromme; punt 4 is die laagste punt; en punte 2 en 6 is op dieselfde hoogte.

- (a) Beskryf die beweging van die deeltjie by $t = 0$?
 (b) Wat is die rigting van beweging by punt 4?
 (c) By watter van die ses genommerde punte draai die deeltjie terug om in die teenoorgestelde rigting te beweeg? Gee rede(s) vir jou antwoord.
 (d) By watter van die ses genommerde punte het die deeltjie die grootste versnelling? Gee rede(s) vir jou antwoord.



$$(a) \text{ Moving in the positive direction – acceleration is positive. } \checkmark \checkmark$$

$$(b) \text{ motion is in the negative direction } \checkmark$$

$$(c) \text{ 3 and 5 } \checkmark - \text{ at this point } v = 0 \text{ m/s and then it moves from +ve to -ve or -ve to +ve } \checkmark$$

$$(d) \text{ 2 and 6 } \checkmark - \text{ at this point the slope of the } v\text{-}t \text{ curve is greatest. } \checkmark$$

Question 5**[1+1+2+4+1+2+2 = 13]**

The position of a particle moving along the x axis depends on the time according to the equation $x = ct^2 - bt^3$, where x is in meters and t in seconds.

(a) What are the units of constant c ?

(b) What are the units of constant b ?

Let their numerical values be 3.0 and 2.0, respectively.

(c) At what time does the particle reach its maximum positive x position?

(d) From $t = 0.0$ s to $t = 4.0$ s, what distance does the particle move?

(e) What is the particle's displacement during the time interval in (d)?

(f) Find its velocity at 4.0 s.

(g) Find its acceleration at 4.0 s.

Die posisie van 'n deeltjie wat langs die x as beweeg, is afhanklik van die tyd, soos aangedui deur die vergelyking $x = ct^2 - bt^3$, waar x in meters en t in sekondes gegee word.

(a) Wat is die eenhede van die konstante c ?

(b) Wat is die eenhede van die konstante b ?

Stel die konstantes se numeriese waardes aan 3.0 en 2.0, onderskeidelik.

(c) Wanneer bereik die deeltjie 'n maximum positiewe x posisie?

(d) Vir die tydsinterval $t = 0.0$ s tot $t = 4.0$ s, wat is die afstand waardeur die deeltjie beweeg?

(e) Wat is die deeltjie se verplasing gedurende die tydsinterval in (d)?

(f) Bereken die snelheid by 4.0 s.

(g) Bereken die versnelling by 4.0 s.

(a) m/s^2 ✓
(b) m/s^3 ✓
$\frac{dx}{dt} = \frac{d}{dt}(3t^2 - 2t^3) = 6t - 6t^2 = 0$
(c) $0 = 6t(1 - t)$ ✓ $t = 0 \text{ s or } t = 1 \text{ s}$ For max value of x , $t = 1 \text{ s}$ ✓
(d) At $t = 0$ $x = 0$ ✓ At $t = 1$ s, $x = 1 \text{ m}$ (Max x value) – velocity is 0 m/s ✓ At $t = 4$ s, $x = 3(16) - 2(64) = -80 \text{ m}$ ✓ Distance travelled: $1 \text{ m} + 1 \text{ m} + 80 \text{ m} = 82 \text{ m}$ ✓
(e) Displacement = $\Delta x = x - x_0 = -80 - 0 = -80 \text{ m}$ ✓
(f) $v = \frac{dx}{dt} = \frac{d}{dt}(3t^2 - 2t^3) = 6t - 6t^2$ ✓ $v(4 \text{ s}) = (6.0 \text{ m/s}^2)(4.0 \text{ s}) - (6.0 \text{ m/s}^3)(4.0 \text{ s})^2 = -72 \text{ m/s}$. ✓
(g) $a = \frac{dv}{dt} = (6 \text{ m/s}^2) - (12 \text{ m/s}^3)t$ ✓ $a(4 \text{ s}) = 6.0 \text{ m/s}^2 - (12.0 \text{ m/s}^3)(4.0 \text{ s}) = -42 \text{ m/s}^2$. ✓

Question 6**[2+2 = 4]**

A key falls from a bridge that is 45 m above the water. It falls directly into a model boat, moving with constant velocity, that is 12 m from the point of impact when the key is released. What is the speed of the boat?

'n Sleutel val van 'n brug af wat 45 m bo die watervlak is. Dit val direk op 'n modelbootjie, wat met 'n konstante snelheid beweeg. Die bootjie was 12 m van die punt waar die sleutel geland het, toe die sleutel losgelaat is. Wat is die snelheid van die bootjie?

Key: $y = -45 \text{ m}$	$a = -9.8 \text{ m/s}^2$	$v_0 = 0 \text{ m/s}$	$t = ?$
Boat: $x = 12 \text{ m}$	$a = 0 \text{ m/s}^2$	$v_0 = ? \text{ m/s}$	$t = ?$
$\Delta y = v_0 t + \frac{1}{2} a t^2$ $-45 = \frac{1}{2} (-9.8) t^2$ ✓✓ $t = \sqrt{\frac{2(-45)}{-9.8}} = 3.03 \text{ s}$			
Time for key to fall			
Also the time for boat to move:			
$\Delta x = v_0 t + \frac{1}{2} a t^2$ $\Delta x = vt$ ✓✓ $v = \frac{\Delta x}{t} = \frac{12}{3.03} = 3.96 \text{ m/s}$			

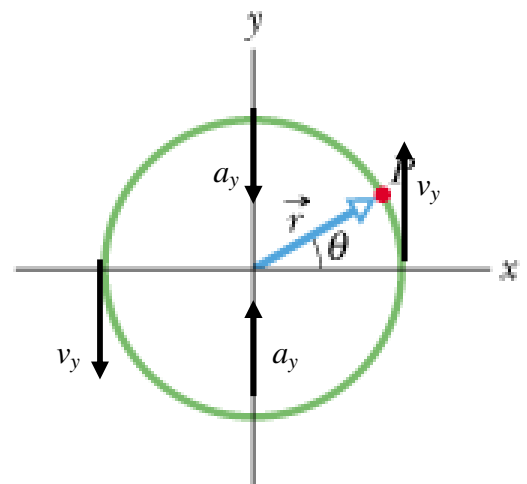
Question 7**[1+2+2 = 5]**

In the Figure particle P is in uniform circular motion, centered on the origin of an xy coordinate system.

- At what values of θ is the vertical component r_y of the position vector greatest in magnitude?
- At what values of θ is the vertical component v_y of the particle's velocity greatest in magnitude? Indicate these velocity components clearly on the Figure.
- At what values of θ is the vertical component a_y of the particle's acceleration greatest in magnitude? Indicate these acceleration components clearly on the Figure.

In die Figuur is die deeltjie P in univormse sikelbeweging, gesentreerd op die oorsprong van die xy koördinaatstelsel..

- By watter waardes van θ het die vertikale component r_y van die posisie vector die grootste waarde?
- By watter waarde van θ het die vertikale component v_y van die deeltjie se snelheid die grootste waarde? Teken hierdie snelheidskomponent duidelik op die Figuur.
- By watter waarde van θ het die vertikale component a_y van die deeltjie se versnelling die grootste waarde? Teken hierdie versnellingskomponent duidelik op die Figuur



- (a) $\theta = 90^\circ$ or 270° ✓

(b) $\theta = 0^\circ$ or 180° ✓ plus 2 vectors on diagram ✓(c) $\theta = 90^\circ$ or 270° ✓ plus 2 vectors on diagram ✓**Question 8****[7]**

A moderate wind accelerates a pebble over a horizontal xy plane with a constant acceleration $\vec{a} = (5,00 \text{ m/s}^2)\hat{i} + (7,00 \text{ m/s}^2)\hat{j}$. At time $t = 0$, the velocity is $(4,00 \text{ m/s})\hat{i}$. What are the magnitude and angle of its velocity when it has been displaced by 12,0 m parallel to the x axis?

'n Redelike sterk wind versnel 'n klippie oor 'n horisontale xy vlak met 'n konstante versnelling $\vec{a} = (5,00 \text{ m/s}^2)\hat{i} + (7,00 \text{ m/s}^2)\hat{j}$. Op tyd $t = 0$, is die snelheid $(4,00 \text{ m/s})\hat{i}$. Wat is die grootte en rigting van die snelheid wanneer die klippie verplaas is met 12,0 m parallel met die x as?

x direction: $\Delta x = 12 \text{ m}$	$v_{0x} = 4.00 \text{ m/s}$	$v_x = ?$	$a_x = 5.00 \text{ m/s}^2$	$t = ?$
y direction: $\Delta y = ?$	$v_{0y} = 0.00 \text{ m/s}$	$v_y = ?$	$a_y = 7.00 \text{ m/s}^2$	$t = ?$
Find t :				
$\Delta x = v_{0x}t + \frac{1}{2}a_x t^2$				
$12 = (4 \text{ m/s})t + \frac{1}{2}(5 \text{ m/s}^2)t^2$ ✓				
$t = \frac{-4 \pm \sqrt{(4)^2 - 4(2.5)(-12)}}{2(2.5)}$ ✓✓				
$t = -3.13 \text{ s}$ or 1.53 s				
Using $t = 1.53 \text{ s}$ $v_x = v_{0x} + at = 4 + (5)1.53 = 11.65 \text{ m/s}$ ✓				
$v_y = v_{0y} + at = 0 + (7)1.53 = 10.71 \text{ m/s}$ ✓				
$v = \sqrt{(11.7)^2 + (10.7)^2} = 15.85 \text{ m/s}$ ✓				
Direction w.r.t. $+x$ axis: $\theta = \tan^{-1} \frac{v_y}{v_x} = \frac{10.7}{11.7} = 42.44^\circ$ ✓				

Question 9**[6 + (3+4+2) = 15]**

9.1 Starting with the equations of motion for free-fall objects, show that the horizontal range of a projectile, R , is given by $R = \frac{v_o^2 \sin 2\theta_0}{g}$, where v_o is the initial velocity, θ_0 is the initial angle of the velocity with the ground and g is the gravitational acceleration 9.8 m/s^2 .

Begin met die bewegingsvergelykings van 'n voorwerp in vryval, en wys dat die horisontale

reikwyte van die projektiel, R , gegee word deur $R = \frac{v_o^2 \sin 2\theta_0}{g}$, waar v_o die aanvanklike

snelheid is, θ_0 die aanvanklike hoek van die snelheid met die grond en g die gravitasie versnelling van 9.8 m/s^2 is.

Let $\Delta x = x - x_0 = R$, and $\Delta y = y - y_0 = 0$. ✓
Then

$$R = (v_{ox} \cos \theta_0) t$$

$$0 = (v_{oy} \sin \theta_0) t - \frac{1}{2} g t^2 \quad \checkmark \checkmark$$

$$\text{Then } t = \frac{2(v_{oy} \sin \theta_0)}{g} \quad \checkmark$$

And

$$R = (v_o \cos \theta_0) \frac{2(v_o \sin \theta_0)}{g} \quad \checkmark$$

$$\text{But } \sin 2\theta = 2 \cos \theta \sin \theta \quad \checkmark$$

Then

$$R = \frac{v_o^2}{g} \sin 2\theta$$

9.2 A ball is shot from the ground into the air. At a height of 9.1 m, its velocity is $\vec{v} = (7.6\hat{i} + 6.1\hat{j})$ m/s,

with $\hat{i}$ horizontal and $\hat{j}$ upward.

(a) To what maximum height does the ball rise?

(b) What total horizontal distance does the ball travel?

(c) What are the magnitude and angle of the ball's velocity just before it hits the ground?

'n Bal word in die lug geskiet van die grond af. Op 'n hoogte van 9.1 m, is die bal se snelheid

$\vec{v} = (7.6\hat{i} + 6.1\hat{j})$ m/s, met $\hat{i}$ horisontaal en $\hat{j}$ opwaarts.

(a) Tot watter maximum hoogte sal die bal styg?

(b) Wat is die totale horisontale afstand wat die bal beweeg?

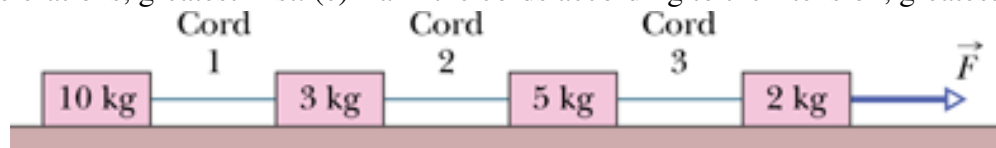
(c) Wat is die grootte en hoek van die bal se snelheid net voor dit die grond tref?

x direction: $x_0 = 0$ m $\Delta x = R$ $v_{ox} = v_x = 7.6$ m/s $a_x = 0$ m/s ²				
y direction: $y_0 = 0$ m $\Delta y = y$ $v_{oy} = ?$ $v_{y, @ \max} = 0$ m/s $a_y = -9.8$ m/s ²				
At $\Delta y = 9.1$ $v_y = 6.1$ m/s				
$v_y^2 = v_{0y}^2 - 2g\Delta y$ Find v_{0y}: $(6.1 \text{ m/s})^2 = v_{0y}^2 - 2(9.8 \text{ m/s}^2)(9.1 \text{ m}) \quad \checkmark$ $v_{0y} = 14.7 \text{ m/s}$				
(a) $v_{y, \max}^2 = v_{0y}^2 - 2g\Delta y_{\max}$ $(0 \text{ m/s})^2 = (14.7 \text{ m/s})^2 - 2(9.8 \text{ m/s}^2)(\Delta y_{\max}) \quad \checkmark \checkmark$ $\Delta y_{\max} = 11.0 \text{ m}$				
(b) $\tan \theta_0 = \frac{14.7}{7.6} \quad \checkmark$ $\theta = 62.6^\circ$ $v_0 = \sqrt{v_{0x}^2 + v_{0y}^2} = \sqrt{(7.6 \text{ m/s})^2 + (14.7 \text{ m/s})^2} \quad \checkmark \checkmark$ $= 16.55 \text{ m/s}$		$R = (v_o \cos \theta_0) t = v_{0x} t$ $0 = (v_o \sin \theta_0) t - \frac{1}{2} g t^2 = v_{oy} t - \frac{1}{2} g t^2$ $t = \frac{2v_{oy}}{g}$ OR $R = \frac{2v_{0x} v_{oy}}{g} \approx 23 \text{ m}$		

$R = \frac{v_0^2}{g} \sin 2\theta = \frac{(16.55)^2}{9.8} \sin 2(62.6^\circ) = 22.8 \text{ m} \checkmark$	
$\tan \theta_0 = \frac{14.7}{7.6}$ below the horizontal (+x axis) $\checkmark \checkmark$	
(c) $v_x = 7.6 \text{ m/s}$, $v_y = 14.7 \text{ m/s}$	$\theta = 62.6^\circ$

Question 10**[5]**

The Figure shows a train of four blocks being pulled across a frictionless floor by force $\vec{F}$. What total mass is accelerated to the right by (a) force $\vec{F}$, (b) cord 3, and (c) cord 1? (d) Rank the blocks according to their accelerations, greatest first. (e) Rank the cords according to their tension, greatest first.



Die Figuur dui vier blokke aan wat getrek word oor 'n wrywinglose oppervlak deur 'n krag deur $\vec{F}$.

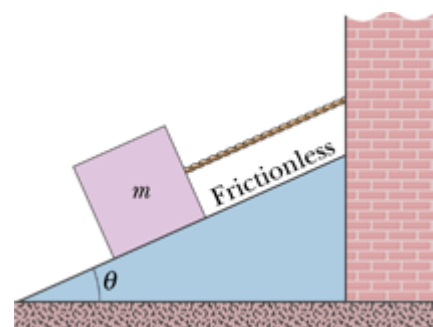
- Wat is die totale massa wat versnel word na regs deur krag $\vec{F}$?
- Wat is die totale massa wat versnel word na regs deur tou 3?
- Wat is die totale massa wat versnel word na regs deur tou 1?
- Skryf die blok massas neer in orde van hulle versnelling, grootste eerste.
- Skryf die tou nommer neer in orde van hulle die spankrag in die tou, grootste eerste.

20 $\checkmark$
18 $\checkmark$
10 $\checkmark$
All tie $\checkmark$
3, 2, 1 $\checkmark$

Question 11**[3+2+2+2 = 9]**

In the Figure, let the mass of the block be 8.5 kg and the angle θ be 30° . Find

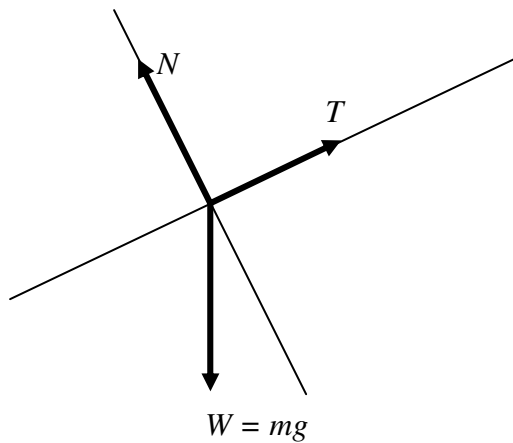
- Draw the free-body diagram of the block.
- Find the tension in the cord.
- Find the normal force acting on the block.
- If the cord is cut, find the magnitude of the resulting acceleration of the block.



Die blok in die figuur het 'n massa van 8.5 kg en die hoek $\theta = 30^\circ$.

- Teken die kragte-diagram van die blok.
- Bepaal die spankrag in die tou.
- Bepaal die normaalkrag wat uitgeoefen word op die blok.
- Indien die tou deurgesny word, bepaal die grootte van die versnelling van die blok.

(a)



✓✓✓

(b)

$$\sum F_x = T - mg \sin \theta = 0$$

$$T = mg \sin \theta$$

$$= (8.5\text{kg})(9.8\text{ m/s}^2) \sin 30^\circ$$

$$= 41.65\text{ N}$$

✓✓

(c)

$$\sum F_y = N - mg \cos \theta = 0$$

$$N = mg \cos \theta$$

$$= (8.5\text{kg})(9.8\text{ m/s}^2) \cos 30^\circ$$

$$= 72.1\text{ N}$$

✓✓

(d)

$$\sum F_x = -mg \sin \theta = ma$$

$$a = -g \sin \theta = -(9.8\text{ m/s}^2) \sin 30^\circ$$

$$= -4.9\text{ m/s}^2$$

✓✓

Question 13**[4]**

Calculate the area between the t axis and the graph of $v = 3 + 4t$ between the points $t = 1$ en $t = 5$. The units of v is in meters per second and that of t is seconds. What are the units of the area which was calculated?

Bereken die oppervlakte tussen die t -as en die grafiek van $v = 3 + 4t$ wat begrens word deur $t = 1$ en $t = 5$. Die eenhede van v is meter per sekonde en die van t is sekonde. Wat is die eenhede van die oppervlakte wat bereken is?

$$\int_1^5 v dt = \int_1^5 (3 + 4t) dt$$

$$= 3t + 2t^2 \Big|_1^5 = (3(5) + 2(5)^2) - (3(1) + 2(1)^2)$$

$$= 60$$

Units of area is meters. ✓

